

CSA0609 - DESIGN AND ANALYSIS OF ALGORITHMS

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DAA LAB EXPERIMENT 2

Common Elements Between Two Arrays

Aim

To write a Python program to find the common elements between two arrays and analyze its time complexity.

Algorithm

1. Start.
2. Read two arrays.
3. Create a set containing the elements of the second array.
4. Traverse the first array.
5. Check whether each element is present in the set.
6. If present, add it to the common elements list.
7. Display the common elements.
8. Stop.

Python Source Code

```
def common_elements(arr1, arr2):  
    set2 = set(arr2)  
    common = []  
    for element in arr1:  
        if element in set2 and element not in common:  
            common.append(element)  
    return common  
  
arr1 = [1, 2, 3, 4, 5, 6]  
arr2 = [4, 5, 6, 7, 8, 9]  
result = common_elements(arr1, arr2)
```

```
print("Array 1:", arr1)
print("Array 2:", arr2)
print("Common Elements:", result)
```

Output



The screenshot shows a Python IDE with a file named 'main.py'. The code defines a function 'common_elements' that takes two arrays, 'arr1' and 'arr2', and returns a list of common elements. The function uses a set to store elements from 'arr2' and iterates through 'arr1' to find common elements. The main code block initializes 'arr1' as [1, 2, 3, 4, 5, 6] and 'arr2' as [4, 5, 6, 7, 8, 9], calls the 'common_elements' function, and prints the results. The output panel on the right shows the execution results: 'Array 1: [1, 2, 3, 4, 5, 6]', 'Array 2: [4, 5, 6, 7, 8, 9]', and 'Common Elements: [4, 5, 6]'.

```
1 def common_elements(arr1, arr2):
2     set2 = set(arr2)
3     common = []
4
5     for element in arr1:
6         if element in set2 and element not in common:
7             common.append(element)
8
9     return common
10
11
12 arr1 = [1, 2, 3, 4, 5, 6]
13 arr2 = [4, 5, 6, 7, 8, 9]
14
15 result = common_elements(arr1, arr2)
16
17 print("Array 1:", arr1)
18 print("Array 2:", arr2)
19 print("Common Elements:", result)
```

Output

```
Array 1: [1, 2, 3, 4, 5, 6]
Array 2: [4, 5, 6, 7, 8, 9]
Common Elements: [4, 5, 6]
```

Result

Thus, the Python program to find the common elements between two arrays was successfully implemented and its complexity was analyzed.